

Version Control

GitHub

What's covered:

1. Creating repositories
2. Authenticating through the terminal

GitHub Creation and Terminal Authentication Guide

Because GitHub no longer accepts normal account passwords when syncing code from your terminal, you must use a **Personal Access Token (PAT)** or the **GitHub CLI** to verify your identity.

1. Authenticating Your Terminal

Option A: Using a Personal Access Token (Classic)

- **Generate Token:** Go to GitHub.com → **Settings** → **Developer Settings** → **Personal Access Tokens** → **Tokens (classic)**.
- **Set Permissions:** Click **Generate new token (classic)** and select the **repo** checkbox scope.
- **Use as Password:** Copy the token. When your terminal prompts you for a password during a `git push`, **paste this token** instead of your account password.

Option B: Using GitHub CLI (Recommended)

- **Install CLI:** Run `brew install gh` (Mac) or `winget install --id GitHub.cli` (Windows).
 - **Login Command:** Run `gh auth login` in your terminal.
 - **Complete Prompts:** Choose `GitHub.com`, select `HTTPS` or `SSH`, and authenticate instantly through your web browser window.
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2. Creating a Repository

Method A: From the GitHub Website

- **Create:** Click the "+" icon in the top right corner of GitHub and select **New repository**.
- **Name & Save:** Enter a repository name, choose **Public** or **Private**, and click **Create repository**.
- **Copy URL:** Copy the HTTPS remote URL provided (e.g., <https://github.com>).

Method B: From the Terminal (Using GitHub CLI)

- **New Repo:** Run `gh repo create repo-name --public` to build a repository on the cloud instantly.
 - **Existing Folder:** Change directories into your existing folder (`cd path/to/folder`) and run `gh repo create`. Follow the interactive prompt to automatically push your local files.
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3. Finalizing the Connection

If you created your repository on the website, link it to your folder using these terminal commands:

- **Rename default branch:** `git branch -M main`
 - **Add remote host:** `git remote add origin https://github.com`
 - **Upload code:** `git push -u origin main`
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